

8 (short) The circuit in Fig. 6 shows a simple form of a 4-bit digital-to-analogue converter. Digital “1” is assumed to be 5 V and digital “0” is assumed to be 0 V.

- (a) Briefly explain the principle of operation. Which of these inputs represent the least significant bit and most significant bit, respectively? [5]
- (b) If the value of R is $1\text{ k}\Omega$ and the operational amplifier is considered to be ideal, determine the input resistance at the terminals S_1 and S_2 . [2]
- (c) Determine the output voltage V_0 if the digital signal 1011 is present at the inputs S_3 S_2 S_1 S_0 . [3]

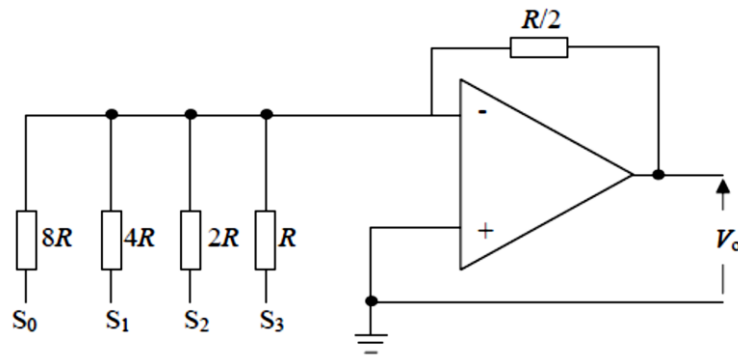


Fig. 6